

EXPERIMENT 1: MATRIX MANIPULATION

1. Concatenation
2. Indexing
3. Randomization
4. Sorting
5. Shifting
6. Reshaping
7. Rotation
8. Flip Function

1. Concatenating-

```
>> A=[2,5,6; 1,4,3; 5,8,4];
```

```
>> display(A);
```

A =

2	5	6
---	---	---

1	4	3
---	---	---

5	8	4
---	---	---

```
>> B=[1,4,6; 4,3,2; 8,7,9]
```

B =

1	4	6
---	---	---

4	3	2
---	---	---

8	7	9
---	---	---

#row concatenation

```
>> C=[A;B];
```

```
>> display(C);
```

C =

```
2 5 6
1 4 3
5 8 4
1 4 6
4 3 2
8 7 9
```

#column concatenation

```
>> D=[A,B];
```

```
>> display(D);
```

D =

```
2 5 6 1 4 6
1 4 3 4 3 2
5 8 4 8 7 9
```

2. Indexing-

```
>> A=[1,2,3,4,5,6];
```

```
>> B=A(5);
```

```
>> display(B);
```

B =

5

```
>> A=[1 2 3;4 5 6];
```

```
>> display(A);
```



A =

1	2	3
4	5	6

```
>> B=A(4);
```

```
>> display(B);
```

B =

5

```
>> A=[1 2;3 4;5 6];
```

```
>> display(A);
```

A =

1	2
3	4
5	6

```
>> B=A(5);
```

```
>> display(B);
```

B =

4

```
>> A=[10 9 4 8 5 1];
```

```
>> B=[3 5];
```

```
>> C=A(B);
```

```
>> display(C);
```

C =



4 5

```
>> A=[1 2 3 4 5 6];
```

```
>> C=A(2:5);
```

```
>> display(C);
```

C =

2 3 4 5

3. Randomization-

```
>> A=rand(4);
```

```
>> display(A);
```

A =

0.8147 0.6324 0.9575 0.9572

0.9058 0.0975 0.9649 0.4854

0.1270 0.2785 0.1576 0.8003

0.9134 0.5469 0.9706 0.1419

```
>> A=randn(4);
```

```
>> display(A);
```

A =

-0.1241 0.6715 0.4889 0.2939

1.4897 -1.2075 1.0347 -0.7873

1.4090 0.7172 0.7269 0.8884



1.4172 1.6302 -0.3034 -1.1471

```
>> A=randi([10 50],1,5);
```

```
>> display(A);
```

A =

21 11 13 43 38

4. Sorting-

```
>> A=rand(3);
```

```
>> B=sort(A);
```

```
>> display(A);
```

A =

0.3171 0.4387 0.7952

0.9502 0.3816 0.1869

0.0344 0.7655 0.4898

```
>> display(B);
```

B =

0.0344 0.3816 0.1869

0.3171 0.4387 0.4898

0.9502 0.7655 0.7952

```
>> C=sort(A,'descend');
```

```
>> display(C);
```



C =

0.9502	0.7655	0.7952
0.3171	0.4387	0.4898
0.0344	0.3816	0.1869

5. Shifting-

```
>> A=[1:5; 6:10; 11:15; 16:20; 21:25];
```

```
>> display(A);
```

A =

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	25

```
>> B=circshift(A,[3,0]);
```

```
>> display(B);
```

B =

11	12	13	14	15
16	17	18	19	20
21	22	23	24	25
1	2	3	4	5
6	7	8	9	10

6. Reshaping-

```
>> A=[1,2,3;4,5,6;7,8,9;10,11,12];
```

```
>> display(A);
```

A =

1	2	3
4	5	6
7	8	9
10	11	12

```
>> B=reshape(A,2,6);
```

```
>> display(B);
```

B =

1	7	2	8	3	9
4	10	5	11	6	12

```
>> C=reshape(A,6,2);
```

```
>> display(C);
```

C =

1	8
4	11
7	3
10	6
2	9
5	12

7. Rotation-

```
>> A=[1,2,3;4,5,6;7,8,9;10,11,12];
```

```
>> display(A);
```

A =

1	2	3
4	5	6
7	8	9
10	11	12

```
>> B=rot90(A);
```

```
>> display(B);
```

B =

3	6	9	12
2	5	8	11
1	4	7	10

```
>> C=rot90(B);
```

```
>> display(C);
```

C =

12	11	10
9	8	7
6	5	4
3	2	1

```
>> D=rot90(C);
```

```
>> display(D);
```

```
D =
```

```
10  7  4  1
```

```
11  8  5  2
```

```
12  9  6  3
```

```
>> E=rot90(D);
```

```
>> display(E);
```

```
E =
```

```
1  2  3
```

```
4  5  6
```

```
7  8  9
```

```
10 11 12
```

```
>> F=rot90(A,3);
```

```
>> G=rot90(A,4);
```

```
>> H=rot90(A,5);
```

```
>> display(F);
```

```
F =
```

```
10  7  4  1
```

```
11  8  5  2
```

```
12  9  6  3
```

```
>> display(G);
```

```
G =
```

```
1  2  3
4  5  6
7  8  9
10 11 12
```

```
>> display(H);
```

```
H =
```

```
3  6  9 12
2  5  8 11
1  4  7 10
```

8. Flip Function-

```
>> A=[1,2,3;4,5,6;7,8,9];
```

```
>> display(A);
```

```
A =
```

```
1  2  3
4  5  6
7  8  9
```

#flip left-right direction

```
>> B=fliplr(A);
```



```
>> display(B);
```

```
B =
```

```
3  2  1
```

```
6  5  4
```

```
9  8  7
```

#flip up-down direction

```
>> C=flipud(A);
```

```
>> display(C);
```

```
C =
```

```
7  8  9
```

```
4  5  6
```

```
1  2  3
```